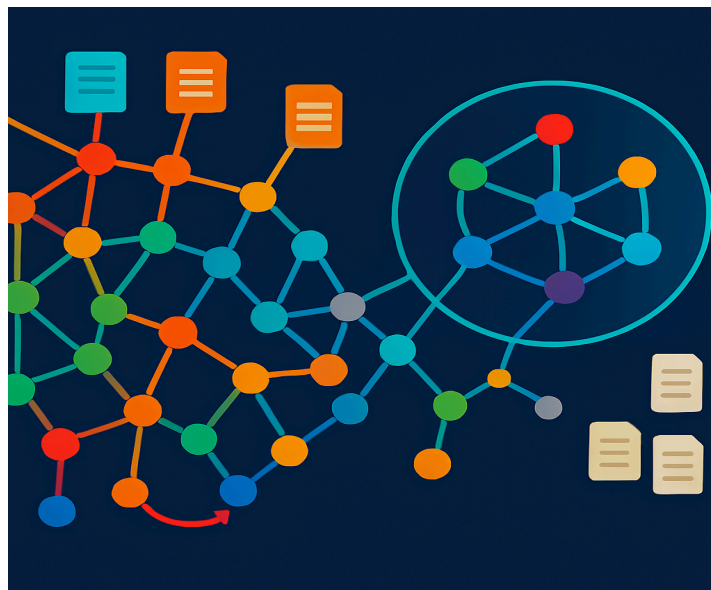




Capstone Project Phase B

25-2-R-2

Skeleton-Based Anomaly Detection in Citation Networks



Omer Goldstein

Ido Sakal

Supervisor: Dr. Dvora Toledano Kitai

[Git](#)

Table Of Contents

1. Introduction.....	1
2. Theoretical Background & Problem Formulation.....	2
2.1. Graphs, Dynamics, and Hyperbolic Embeddings.....	3
2.2 Preliminaries.....	5
2.2.1 Poincaré Space.....	5
2.2.2 The DiskNet Model.....	6
2.2.3 Anomaly Detection.....	8
2.3 Problem Formalization.....	8
3. Research Process and System Implementation.....	9
3.1 Approach.....	9
3.1.1 Stage 0: Node Representation in Hyperbolic Space.....	9
3.1.2 Stage 1: Learning Dynamic Representations in Poincaré Space.....	10
3.1.3 Stage 2: Forecasting Future Positions in Hyperbolic Space.....	11
3.1.4 Stage 3: Reconstruction via Lifting Operation.....	11
3.1.5 Stage 4: Anomaly Identification Through Comparative Analysis.....	11
3.1.6 Stage 5: Anomaly Classification.....	11
3.1.7 Validation of Anomaly Detection Module.....	12
3.2 Research Process.....	14
3.3 System Implementation: Codebase, Language & Libraries.....	15
3.4 Challenges During Implementation.....	16
4. Results and Conclusions.....	17
4.1 Results and Evaluation.....	17
4.2 Conclusions and Future Work.....	20
References.....	21
Appendix.....	22
User Guide: DiskNet-Based Anomaly Detection System.....	23
Maintenance Guide: DiskNet for Citation Networks.....	31

Abstract

Citation networks are dynamic systems that evolve as new academic works are published and referenced. These networks exhibit hierarchical structure, scale-free degree distributions, and temporally heterogeneous behaviors, which traditional static models often fail to capture. To address this, we propose a geometry-aware and temporally sensitive framework that combines hyperbolic graph embeddings with continuous-time neural dynamics for the analysis of evolving citation networks.

The proposed method builds on the DiskNet framework, which combines Poincaré embeddings and Neural ODEs to model network dynamics. Node embeddings are initialized using latent geometry embedding techniques, projected into hyperbolic space, and evolved over time. The resulting node positions are used to reconstruct a predicted citation graph, which is compared to the actual graph to compute deviation scores. These scores are used for anomaly detection via unsupervised methods like Isolation Forest.

The model focuses on identifying anomalous nodes whose citation behavior deviates from expected dynamics. Experimental validation on real-world citation datasets aims to demonstrate the framework's effectiveness in capturing structural organization and temporal evolution, offering a foundation for future research in dynamic network modeling, anomaly detection, and scholarly influence analysis.

1. Introduction

The rapid growth of complex networked-structured data in scientific domains has prompted the development of sophisticated models that can capture both the intricate structural and temporal dynamics inherent in such systems. Among these, citation networks represent a particularly rich and dynamic form of scholarly communication, where nodes correspond to academic papers and directed edges denote citation relationships. These networks not only encode the structural interdependence of scientific knowledge but also evolve continuously as new papers are published and cited, often forming hierarchical, scale-free structures that encode both academic influence and domain-specific knowledge flows.

Traditional approaches to network analysis have often relied on static representations, which assume a fixed set of nodes and edges. While effective in revealing structural properties at a specific moment, these models inherently fail to capture temporal phenomena such as the rise of influential works, the formation of new research communities, or the emergence of anomalous citation behaviors. Consequently, dynamic graph modeling has emerged as a critical research area aimed at addressing these limitations. By incorporating temporal information directly

into the modeling process, dynamic graph frameworks enable a more expressive understanding of network changes over time.

Recent advances in graph representation learning have further enriched this approach through graph embedding techniques, which transform high-dimensional, structured data into low-dimensional vector spaces while preserving both topological and semantic properties. Such embeddings enable anomaly detection by highlighting deviations from typical structural patterns.

This work focuses on the modeling of citation networks as dynamic systems embedded in non-Euclidean, specifically hyperbolic, geometric spaces. The inherent hierarchical structure and power-law degree distributions of citation networks render them well-suited for representation in negatively curved spaces, where hyperbolic embeddings can more naturally encode these relationships. The Poincaré disk model offers a computationally tractable framework for such embeddings.

To capture temporal dynamics in this geometry, we employ Neural Ordinary Differential Equations (Neural ODEs), allowing us to model the continuous-time evolution of node embeddings. This methodology enables the forecasting of future states and the detection of anomalous patterns by comparing predicted and observed network configurations. The primary modeling framework is the Dynamics-Invariant Skeleton Neural Network (DiskNet), which integrates hyperbolic embeddings and neural ODEs to learn a low-dimensional skeleton. Deviations between predictions and actual behavior identify anomalous citation patterns.

This project explores the integration of hyperbolic geometry and temporal modeling to address anomaly detection in evolving citation networks. By integrating these methodologies, the work contributes to the intersection of machine learning, network science, and geometric representation learning, offering both theoretical insights and practical methodologies for the analysis of complex dynamic systems.

The remainder of this document is organized as follows. Section 2 establishes the theoretical foundations underlying dynamic graph analysis and introduces the core geometric components of the DiskNet framework, concluding with the formal problem formulation. Section 3 presents the research process and system implementation, detailing the proposed methodology for anomaly detection, the technical infrastructure, and the challenges encountered during development. Finally, Section 4 reports the empirical evaluation results and discusses the main conclusions and some optional future research directions.

2. Theoretical Background & Problem Formulation

2.1. Graphs, Dynamics, and Hyperbolic Embeddings

Graph-based data structures, commonly referred to as graph networks, offer a powerful and flexible framework for modeling complex systems composed of discrete entities and their relationships. Formally, a graph $G = (V, E)$ consists of a set of nodes V , representing individual entities (e.g., academic papers), and a set of edges E , which encodes relationships or interactions between them (e.g., citation between papers). This abstraction is widely used to model complex systems across diverse domains, including social systems, biological networks, and scholarly communication, with networks evolving as new connections form or dissolving over time.

In practical applications, nodes in a graph are typically associated with feature vectors; a structured, fixed-length representation capturing heterogeneous information relevant to the modeled system. These features may reflect structural properties (e.g., degree centrality or clustering coefficient), semantic attributes derived from content (such as keywords, abstracts, or topics extracted from the text), and temporal metadata describing the evolution of each entity (e.g., publication year). Feature-based modeling serves as input to a wide range of machine learning techniques for network-related tasks such as node classification, link prediction, and anomaly detection.

Many real-world networks are inherently dynamic, with nodes and edges appearing, disappearing, or changing over time. Dynamic graph models extend static formulations by explicitly incorporating temporal evolution, allowing the analysis of non-stationary behaviors and long-term structural trends. This temporal perspective is particularly important for detecting anomalies that may emerge gradually or manifest as abrupt deviations from expected structural evolution.

Citation networks [1] constitute a prominent example of such dynamic graph systems. In citation networks, nodes correspond to academic publications and edges denote citations from one paper to another. These networks are acyclic, temporally ordered, and highly heterogeneous, reflecting the cumulative and hierarchical nature of knowledge production. As new publications enter the system and citation patterns evolve, citation networks naturally exhibit scale-free degree distributions and community-driven organizations. Modeling citation networks as dynamic systems enables more comprehensive analyses, including the identification of anomalous citation behaviors and the characterization of scholarly influence over time.

To facilitate computational analysis of graph-structured data, graph embedding methods aim to encode nodes into low-dimensional vector spaces while preserving essential structural and semantic properties of the original network. Machine learning and deep learning-based approaches [5], such as Graph Convolutional Networks (GCN's), leverage neural architectures to integrate node features with local graph

topology through differentiable optimization. While these methods have demonstrated strong performance in static or moderately sized graphs, their Euclidean formulations often struggle to faithfully represent networks characterized by hierarchical organization and heavy-tailed connectivity patterns.

Dynamic graph embedding methods [3] extend static embeddings by incorporating temporal evolution patterns, enabling node representations that reflect both structural dependencies and temporal dynamics. Unlike static embeddings, which assume a fixed graph topology, dynamic approaches model evolving node relationships over time. These methods often rely on continuous-time formulations or temporal aggregation mechanisms to capture temporal dynamics at arbitrary time intervals rather than being restricted to discrete snapshots. A central challenge lies in preserving both local structural properties while simultaneously modeling temporal dynamics, particularly in networks characterized by hierarchical or scale-free structures.

Latent geometry embedding methods address this limitation by assuming that observed network structure arises from an underlying geometric space [2]. Rather than imposing a predefined topology, these methods infer hidden geometric coordinates such that observed connections correspond to proximity in the latent space. Hyperbolic geometry, in particular, has emerged as a natural choice for modeling complex networks due to its constant negative curvature and exponential volume growth, which enable low-distortion representations of hierarchical and scale-free structure.

The Mercator algorithm [2] represents a paradigmatic implementation of this approach, specifically designed to embed networks into hyperbolic space through maximum likelihood estimation. By jointly inferring node positions, hidden degrees, and global model parameters, Mercator provides an efficient and principled mechanism for uncovering latent geometric structure from network topology. Such hyperbolic embeddings support a wide range of downstream tasks, including community detection, link prediction, and dynamic network analysis.

Within the context of dynamic citation networks offer a compact and expressive representation that naturally encodes both similarity and hierarchical influence. Their geometric properties make them particularly well-suited for modeling the long-term evolution of scholarly communication and for identifying deviations from expected structural patterns. These characteristics form the conceptual foundation for the hyperbolic and dynamic modeling framework adopted in this study and motivate the integration of geometric learning with temporal dynamics in the proposed anomaly detection approach.

2.2 Preliminaries

2.2.1 Poincaré Space

Hyperbolic geometry is a non-Euclidean geometric space characterized by constant negative curvature. Unlike Euclidean geometry, in which the sum of angles in a triangle equals 180 degrees and parallel lines remain equidistant, hyperbolic geometry permits the divergence of parallel lines and results in a triangle angle sum less than 180 degrees. These properties lead to exponential volume growth as a function of distance from a reference point, enabling the embedding of hierarchical structures with significantly lower distortion than Euclidean space.

Negative curvature induces a metric where distances grow exponentially with radius rather than linearly, making hyperbolic space particularly suitable for representing complex networks with power-law degree distributions. Such properties are commonly observed in real-world systems such as social networks, brain connectivity networks, and infrastructure systems [4].

The Poincaré disk model provides a conformal representation of hyperbolic geometry within a bounded domain. More generally, it is defined as an n -dimensional ball embedded in Euclidean space, centered at the origin, with radius $\frac{1}{\sqrt{|c|}}$ where $c < 0$ denotes the curvature parameter of the hyperbolic space (typically set to $c = -1$). Formally, the space is defined as:

$$(1) H^{n,c} = \left\{ x \in R^n \mid ||x|| < \frac{1}{\sqrt{|c|}} \right\}$$

The hyperbolic distance between two points x and y in the Poincaré ball is defined by:

$$(2) d_H^c(x, y) = \frac{1}{\sqrt{|c|}} \operatorname{arcosh} \left(1 - \frac{2c||x-y||^2}{(1+c||x||^2)(1+c||y||^2)} \right), \quad c < 0$$

By representing nodes in hyperbolic space, the model can effectively encode complex topological features and temporal dynamics, providing a compact and expressive representation for tasks such as anomaly detection and prediction of evolving citation patterns.

Hyperbolic embeddings are optimized to preserve distances and hierarchical relationships, by leveraging logarithmic maps that project points from the curved manifold onto a local Euclidean tangent space. This projection enables stable distance computation and gradient-based optimization while explicitly accounting for the underlying curvature of the space.

2.2.2 The DiskNet Model

The **Dynamics-Invariant Skeleton Neural Network (DiskNet)** [4] is a deep learning framework designed to model long-term dynamics in complex networks. By integrating hyperbolic embeddings with **Neural Ordinary Differential Equations (ODEs)**, DiskNet predicts future states of dynamic systems through a compact, structure-preserving representation (termed a “skeleton”) of the original graph. This skeleton captures essential topological and dynamical features, enabling efficient and accurate modeling. A central challenge in modeling network dynamics is identifying a low-dimensional representation that preserves the network’s intrinsic behavior. DiskNet addresses this via the **Hyperbolic Renormalization Group (RG) module**, which maps nodes to super-nodes while maintaining topological similarity and dynamical fidelity. Initial hyperbolic embeddings are obtained using the **Mercator algorithm** [2], which assigns angular and radial coordinates in the Poincaré disk to preserve hierarchical structure and node similarity. Embeddings are projected to the tangent space via the logarithmic map for distance computations.

The assignment matrix P mapping nodes to super-nodes is computed as:

$$(3) \quad P = \text{softmax}(\tilde{C}_s \tilde{C}^T)$$

where $\tilde{C}$ and $\tilde{C}_s$ are transformed representations of original and super-nodes, respectively. The process is regularized by one-hot loss and near-discrete assignments and reconstruction loss to ensure discrete assignments and topological consistency.

Once the skeleton is formed, node states are aggregated to super-nodes using normalized Laplacian convolutions:

$$(4) \quad X_s = PH,$$

where H is the representation of original nodes obtained through normalized Laplacian operations and MLP transformations. This aggregation enables dynamic simulation on the coarse skeleton while retaining key behavioral patterns of the original system.

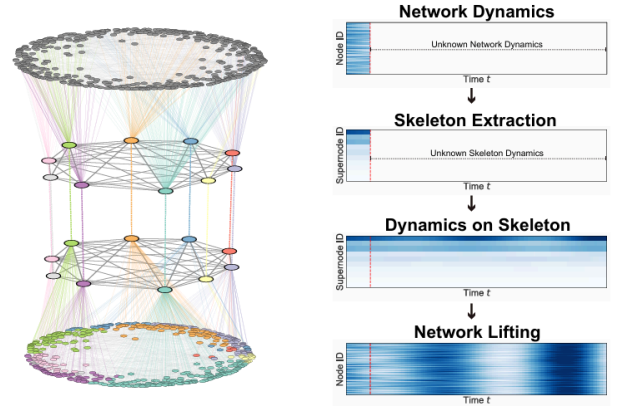


Figure 1: Long-term prediction via the skeleton of complex network dynamics. [4]

The temporal evolution of the skeleton is modeled through Neural ODEs:

$$(5) \quad \frac{dZ_s}{dt} = f(Z_s) + g(Z_s, A_s)$$

where Z_s are the latent super-nodes states, and A_s is the skeleton adjacency matrix, f (self-dynamics) captures self-dynamics via a Multi-Layer Perceptron (MLP), and g represents interaction dynamics with neighbors. is modeled via graph operations. Solving these ODEs predicts skeleton trajectories over arbitrary time horizons.

Finally, super-node trajectories are lifted back to the original graph to recover node-level dynamics using a degree-based clustering super-resolution module. Nodes are clustered via K-means on the logarithm of their degrees, refining predictions based on structural similarity and academic impact patterns.

This approach enables long-term prediction and anomaly detection in evolving citation networks, while reducing computational complexity by operating on a compact skeleton representation.

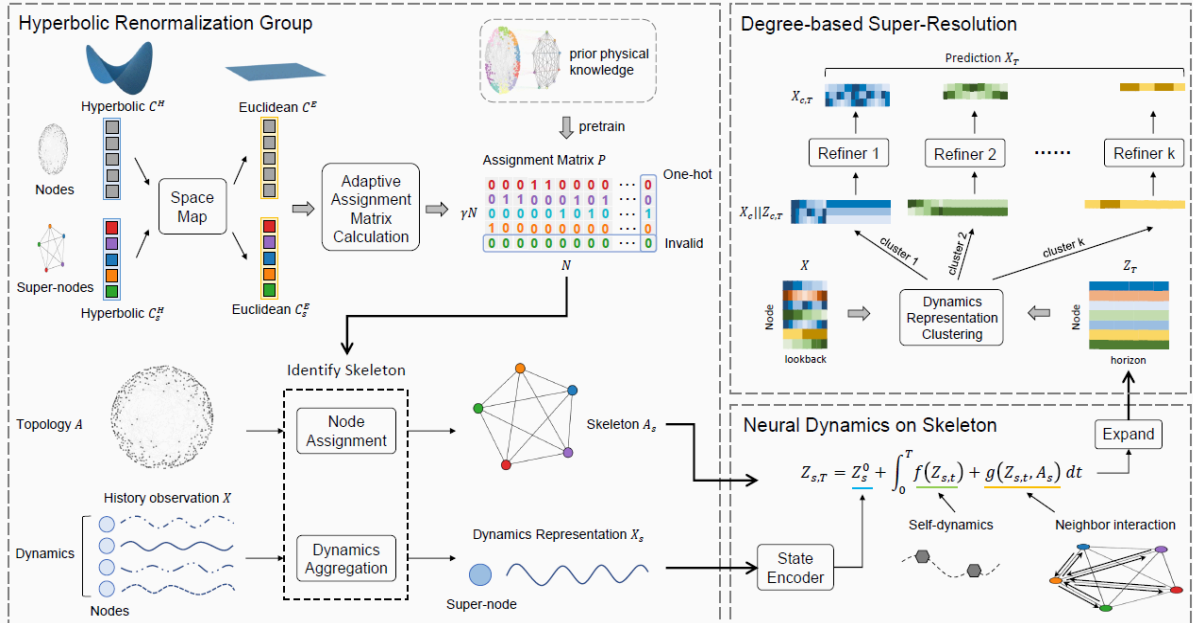


Figure 2: Overall framework of DiskNet: (1) Hyperbolic Renormalization Group, which identifies the representation and skeleton of network dynamics; (2) Neural Dynamics on Skeleton, which models the dynamics of super-nodes on the skeleton; and (3) Degree-based Super-Resolution, which lifts the predicted values of super-nodes to the original nodes. [4]

2.2.3 Anomaly Detection

Dynamic complex networks, such as citation systems, continuously evolve as new publications and citation links emerge. These networks are characterized by high dimensionality, heavy-tailed degree distributions, and the spontaneous formation of community structures, which collectively pose significant challenges for effective anomaly detection.

Citation networks, in particular, are directed, acyclic, and inherently heterogeneous. As such, anomaly detection methods often fail to capture structurally subtle yet semantically anomalous citations patterns, such as cross-disciplinary citations or links between otherwise disconnected scientific domains. Skeleton-based approaches aim to reveal these latent network dynamics and interdependencies by mapping the complex graph into a compact and interpretable representation.

Isolation Forest is an efficient approach for anomaly detection based on the principle of recursively partitioning data using isolation trees. where, data points are split by randomly selecting features and corresponding thresholds. The core of this mechanism relies on the observation that anomalous instances require fewer splits to be isolated compared to normal points, as they typically occupy sparse regions in the feature space.

An anomaly score is calculated based on the normalized average path length required to isolate each point across all trees. Higher scores indicate a greater likelihood of anomalous behavior, providing an effective threshold-based classification without requiring labeled data or assumptions about normal behavior patterns. In the context of citation networks, deviation scores from predicted behaviors are used, enabling detections of patterns that deviate from typical citation dynamics. This approach is particularly suitable for dynamic networks where anomalies are subtle and evolve over time.

2.3 Problem Formalization

The proposed approach for detecting anomalies in academic citation networks builds upon the DiskNet framework and leverages the intrinsic properties of hyperbolic geometry to effectively model dynamic graph structures over time. This section outlines the comprehensive methodology for identifying citation anomalies through a multi-stage pipeline.

We formally define the citation network as a directed graph $G = (V, E)$, where V represents the set of nodes (academic papers), and E denotes the set of directed edges (citation relationships). While citation networks inherently evolve over time, our approach models temporal dynamics by learning continuous latent trajectories from historical feature observations on a fixed static network structure using Neural

Ordinary Differential Equations (Neural ODEs).

This formulation enables the capture of evolutionary citation patterns while accommodating the practical constraints of working with static citation datasets. The temporal resolution of the learned dynamics is configurable based on modeling requirements, enabling analysis across multiple time horizons and supporting the identification of both short-term citation bursts and long-term influence propagation.

The primary objective is to identify anomalous nodes, papers that exhibit unexpected citation behavior deviating significantly from the network's predicted evolutionary trajectory. Such anomalies may include papers that experience sudden citation surges, unexpected cross-disciplinary citations, or citation patterns that contradict established academic influence flows.

3. Research Process and System Implementation

3.1 Approach

System Objectives and Implementation Architecture

The proposed system addresses anomaly detection in dynamic citation networks by developing a framework that models how citation patterns change over time and identifies unusual behaviors. The main goal is to create a system capable of predicting future citation behavior and detecting papers that exhibit abnormal patterns compared to standard academic practices. The system adopts a multi-stage approach: representing papers as nodes, modeling citations evolution over time, predicting future network states, and comparing predictions with observed data to identify anomalous behaviors, such as unexpected citation bursts or atypical cross-field references.

The target users include researchers studying citation patterns, universities tracking research trends, and digital libraries requiring automated tools to detect irregular citation behavior. The framework supports researchers in understanding how academic influence propagates over time and provides tools for analyzing knowledge flow in scientific networks. The implementation is designed to be accessible for academic research while maintaining the ability to analyze large-scale citation datasets.

3.1.1 Stage 0: Node Representation in Hyperbolic Space

To characterize citation behavior patterns, each node is augmented with three domain-specific features designed to capture potentially anomalous citation dynamics:

- **Topic Diversity** quantifies the interdisciplinary spread of received citations within a temporal window. It is defined as the ratio of unique research topics among citing papers to total number of citations, with lower diversity potentially indicating suspicious citation clustering within narrow academic domains.
- **Neighborhood Inertia** evaluates the academic impact of citing papers based on their cumulative citation counts, computed as:

$$(6) \frac{1}{\text{average neighbor impact} + 1}.$$

Higher values indicate citations originating from low-impact or potentially fictitious publications, as the inverse relationship emphasizes papers cited predominantly by academically weak sources.

- **Citer-to-Cited Distance** measures the average shortest-path length between citing and cited papers in the existing network, excluding the citation edge under consideration. A penalty distance of 10.0 is assigned when no path exists. Larger distances may indicate either interdisciplinary breakthroughs or anomalous citation patterns from academically distant domains.

These features are then integrated into the hyperbolic embedding process. Initial hyperbolic coordinates are generated using the Mercator algorithm, which captures both homophily (i.e., similar nodes tend to be connected) and structural equivalence (nodes with similar network roles).

The resulting hyperbolic coordinates are represented in the Poincaré disk, leveraging the natural ability of hyperbolic geometry to encode hierarchical structures with low distortion.

3.1.2 Stage 1: Learning Dynamic Representations in Poincaré Space

At this stage, the DiskNet model is employed to learn latent temporal dynamics from the static citation network to capture both spatial and temporal patterns in hyperbolic space. For each node v during historical evolution, the model learns:

- A hyperbolic position vector $z_v^t \in H^d$, representing its location in the Poincaré disk.
- A continuous-time temporal evolution model defined via a Neural ODEs, describing how z_v^t evolves over time.

This dual representation captures both the static structural role of each paper within the citation network and its observed temporal evolution.

3.1.3 Stage 2: Forecasting Future Positions in Hyperbolic Space

Using the learned Neural ODE, the future feature state x_v^{t+1} of each node is predicted, reflecting its expected position in the latent hyperbolic space at the next time step $t + 1$. These predictions reflect the model's expectation regarding how each paper's influence and connectivity will naturally evolve based on historical patterns and underlying network structure.

3.1.4 Stage 3: Reconstruction via Lifting Operation

The forecasted skeleton-level dynamics are then mapped back to the original network through a reconstruction (lifting) operation.

Predicted supernode states are projected back to individual nodes using the learned assignment matrix, producing a forecasted feature vector for each paper. This yields a predicted set of node features that represents the expected citation state at the next time-step.

3.1.5 Stage 4: Anomaly Identification Through Comparative Analysis

Anomalies are identified by comparing predicted and observed citation dynamics :

- For each node v , a deviation score is computed, quantifying the discrepancy between predicted and actual behavior.

Specifically, the Mean Absolute Error (MAE) between the predicted and observed feature vectors is used.

The deviation score provides a quantitative measure of how significantly a paper's actual citation behavior diverges from its forecasted trajectory.

3.1.6 Stage 5: Anomaly Classification

In the final stage, nodes are classified as either normal or anomalous based on their deviation scores:

1. A node is considered anomalous if its score exceeds a predefined threshold, indicating a substantial deviation from expected citation behavior.
2. Threshold selection is performed using one of the following strategies:
 - Unsupervised Statistical methods, such as Isolation Forest, were applied to deviation scores.
 - Iterative validation and cross-run aggregation, where the pipeline is executed over k independent runs. This allows for the calculation of mean deviation scores and the application of dual thresholds based on standard deviation to identify nodes that consistently exhibit high anomaly scores across iterations.

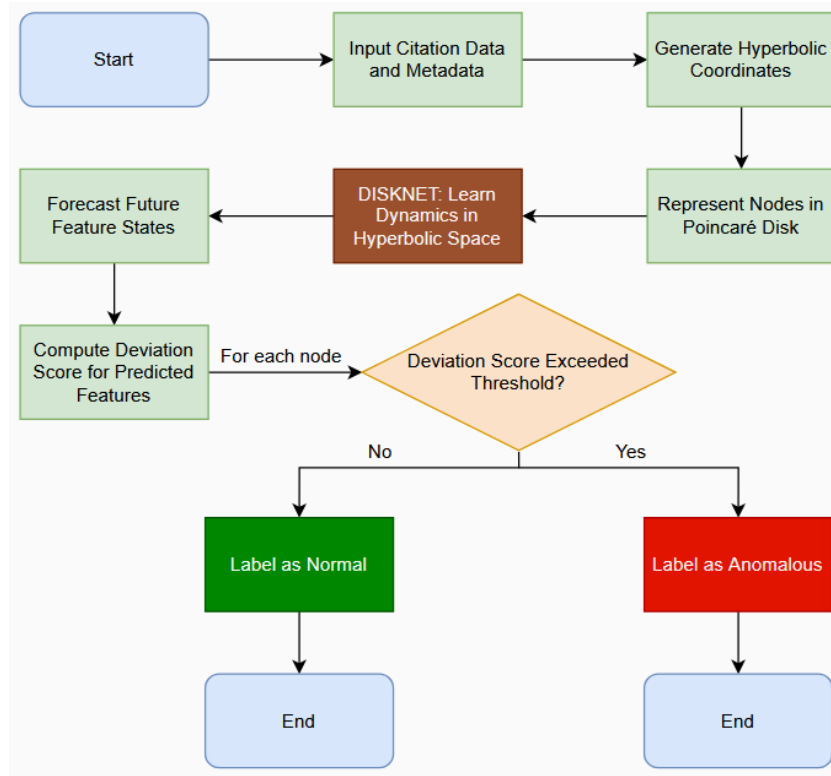


Figure 3: Anomaly Detection Pipeline in Dynamic Citation Networks, Stages 0-5

3.1.7 Validation of Anomaly Detection Module

To evaluate the performance and robustness of the proposed anomaly detection framework, we design a controlled validation procedure based on synthetic anomaly injections and iterative testing. This approach enables systematic assessment of the model's ability to detect anomalous patterns under varied and reproducible conditions.

Synthetic Anomaly Injection

To simulate realistic yet controlled anomalies, a clean citation network is modified as follows:

1. Artificial nodes injection: 10% additional synthetic nodes are added to the graph, representing fictitious papers that are absent from the original dataset.
2. Random citation behavior: Synthetic edges amounting to 30% of the original edge count are added. Each synthetic node exclusively cites older, existing papers, simulating atypical citation patterns, such as unexpected cross-domain patterns or irregular citation bursts of citations.

This procedure introduces localized noise and anomalies without significantly altering the global network structure, thus allowing for targeted evaluation of the detection module.

Iterative Anomaly Detection

To assess stability and consistency, the anomaly detection pipeline is executed over K independent iterations. In each iteration:

1. A new random sample of artificial nodes and anomalous edges is injected into the graph.
2. The full detection pipeline is applied, including feature extraction, hyperbolic projection, temporal modeling via DiskNet, and deviation-based anomaly scoring.
3. Detected anomalies are recorded and aggregated across iterations to assess consistency and generalization and to filter stochastic noise from extreme deviation scores.

Evaluation Metric - Recall

Recall measures the proportion of injected anomalies correctly identified:

$$(7) \text{ Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

- High Recall (close to 1) indicates that the model successfully detects most of the artificially injected anomalies, demonstrating sensitivity to abnormal citation behavior.
- Low Recall (close to 0) suggests that many anomalies go undetected, potentially due to underfitting or insufficient modeling of dynamic patterns.

By tracking the Recall score, we measure the model's ability to catch true anomalies.

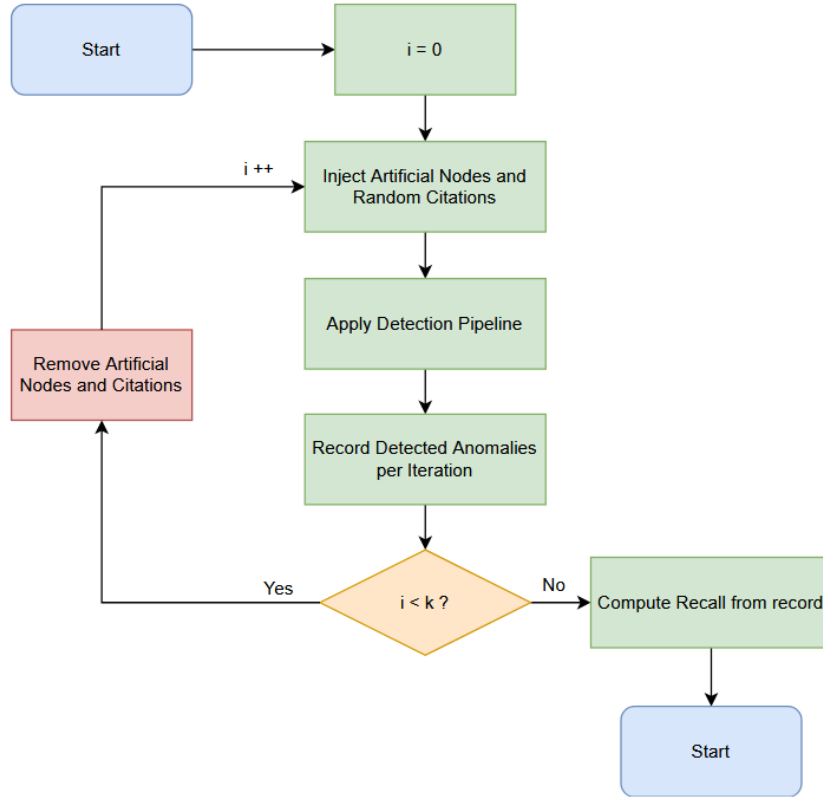


Figure 4: Validation process pipeline, step 3.1.7

3.2 Research Process

We conducted a literature review to understand methods for dynamic graph analysis and anomaly detection in citation networks. The review focused on the use of hyperbolic geometry for modeling hierarchical network structures and on Neural Ordinary Differential Equations for capturing continuous-time dynamics. These studies guided the selection of the methodological components and their integration into the proposed citation anomaly detection framework.

System Implementation

The original DiskNet framework code was obtained from a public GitHub repository and adapted for citation network analysis. The main modifications included adjusting the original dynamic modeling components to fit citation-specific behavior, modifying the data loading pipeline to support citation datasets, and incorporating an anomaly detection mechanism based on deviation scoring. In addition, data preprocessing procedures were implemented to clean and format citation data. An Isolation Forest module was used for anomaly classification, and the complete system was tested to ensure correct interaction between all components.

3.3 System Implementation: Codebase, Language & Libraries

The implementation of this project is based on the Python programming ecosystem, which provides a mature and versatile environment for scientific computing, machine learning, and graph -based modeling. Python's readability, extensive community support, and extensive range of libraries make it particularly well-suited for experimentation, rapid prototyping, and deep learning workflows.

Programming Language

Python (version 3.10 or higher). Python offers robust tools for data manipulation, numerical computation, and model development. Its ecosystem supports advanced neural modeling and graph processing, making it suitable for implementing the proposed dynamic citation network pipeline.

Core Libraries and Their Roles

PyTorch: Deep learning framework used for implementing Neural ODEs.

torchdiffeq: Enables numerical solvers for Neural Ordinary Differential Equations.

NetworkX: Constructing, analyzing, and manipulating citation graph structures.

scikit-learn: Anomaly detection using Isolation Forest and evaluation metrics.

Mercator: Provides hyperbolic embeddings via maximum likelihood estimation in the Poincaré disk model.

NumPy / SciPy: Provides efficient tools for linear algebra and matrix operations.

Matplotlib / scienceplots: Visualizations of graph dynamics and embedding trajectories.

Requests / OmegaConf: HTTP requests for data downloading and YAML configuration management.

Dataset

The implementation utilizes a citation dataset derived from OpenAlex, containing academic papers and their citation relationships across multiple research domains, including Computer Science, Genetics, Biology, Mathematics, Environmental Science, Physics, and Sociology. The original OpenAlex dataset was adapted and preprocessed for this study, including temporal consistency corrections, data quality filtering, and domain-specific feature extraction, to ensure compatibility with the proposed anomaly detection framework.

Runtime Environment

The project is executed in a Google Colab environment to support efficient training, evaluation, and experimentation. The platform provides sufficient computational resources for hyperbolic embedding operations and Neural ODE training. Additionally, the platform's accessibility and integration with Python-based machine

learning frameworks (e.g., PyTorch, scikit-learn) ensure compatibility and ease of development throughout all phases of this research project.

3.4 Challenges During Implementation

1. Adaptation of the DiskNet Framework

The original **DiskNet** codebase was designed to model physical dynamics systems, such as power grids and biological networks, which differ from citation networks in both structure and semantics. The framework assumes continuous physical dynamics governed by differential equations, while citation networks evolve through discrete events (publications and citation links) that accumulate over time. Adapting the framework, therefore, required modifications across multiple components of the system.

Data Pipeline Restructuring

The data pipeline was restructured to accommodate citation networks characterized by directed edges and temporal metadata. While the original model expected undirected graphs with node-level time series representing continuous physical measurements, the adapted pipeline had to support directed citation relationships, where edges correspond to one-time events rather than ongoing interactions.

Shifting from State Prediction to Anomaly Detection

The core objective of the model was redirected from predicting future node states toward identifying anomalous behavior. Whereas the original DiskNet framework optimizes forecasting node trajectories in the latent space, our adaptation focuses on identifying nodes whose observed trajectories deviate significantly from predicted patterns. This shift required introducing deviation scoring mechanisms and modifying the output interpretation to support anomaly classification rather than trajectory accuracy measurement.

Evaluation Metrics

New evaluation routines were developed to measure anomaly detection performance, replacing the original prediction-oriented metrics. While the original framework relied primarily on Mean Absolute Error (MAE) between predicted and observed node states, our adaptation settings required recall and precision-based metrics suitable for highly imbalanced classification scenarios where anomalies are rare.

Network Size Generalization

The architecture was generalized to support citation networks of varying sizes. The original implementation assumed fixed network dimensions determined at training time. We modified the model to dynamically adjust its internal structures based on input network size, allowing a single trained model to process different citation networks of different scales without retraining.

2. Feature Selection and Engineering

Selecting effective node-level features for detecting anomalous citation behavior proved challenging due to inherent tradeoffs. Several features that seemed promising in theory failed to generalize well in practice, either by incorrectly flagging legitimate papers as anomalous or by missing subtle manipulation patterns.

Through iterative experimentation, we converged on the three features described in stage 0 in subsection 3.1. These features demonstrated consistent discriminative performance across different network configurations and generalized well to unseen data.

3. Dataset Preprocessing

The citation dataset derived from OpenAlex required careful preprocessing prior to use. Self-citations were identified and removed to prevent artificial inflation of citation counts, and entries corresponding to non-academic content (e.g., conference abstracts or editorial notes) were filtered out to ensure that the dataset accurately represented genuine scholarly work.

4. Results and Conclusions

This section presents the empirical results obtained from applying the proposed anomaly detection framework on real-world citation data. To evaluate the model's performance, with a particular focus on its ability to identify controlled synthetic anomalies injected into the citation network, as described in the validation protocol.

4.1 Results and Evaluation

Dataset Configuration and Experimental Setup

The experimental evaluation was conducted on a curated subset of the OpenAlex citation dataset. Following the synthetic anomaly injection strategy detailed in Section 3.1.7, the validation network was constructed by augmenting a base citation graph containing approximately 6,250 original academic papers with an additional 10% synthetic nodes (approximately 625 nodes).

These synthetic nodes are designed to simulate anomalous citation behavior, including irregular citation patterns and atypical connections to existing papers. As a result, the final validation network comprised nearly 7,000 nodes along with their corresponding citation links.

The anomaly detection pipeline was configured with $k=10$ independent iterations. Empirical testing demonstrated that this number of iterations provided stable and reliable performance, with no significant improvement in detection accuracy observed at higher iteration counts. Consequently, $k = 10$ was selected as a trade-off between computational efficiency and result stability.

Results

To understand how the model perceives and classifies nodes, we examine the visual outputs from Iteration 7:

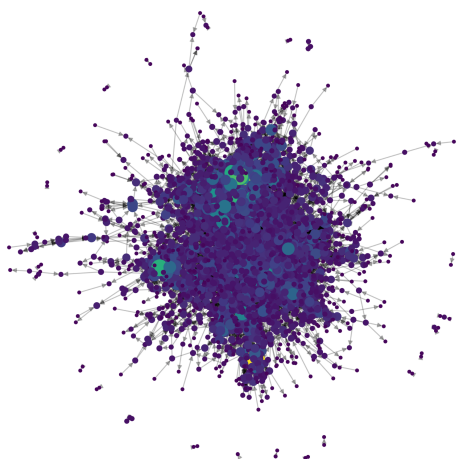


Figure 5: Citation Network Visualization (Iteration 7)

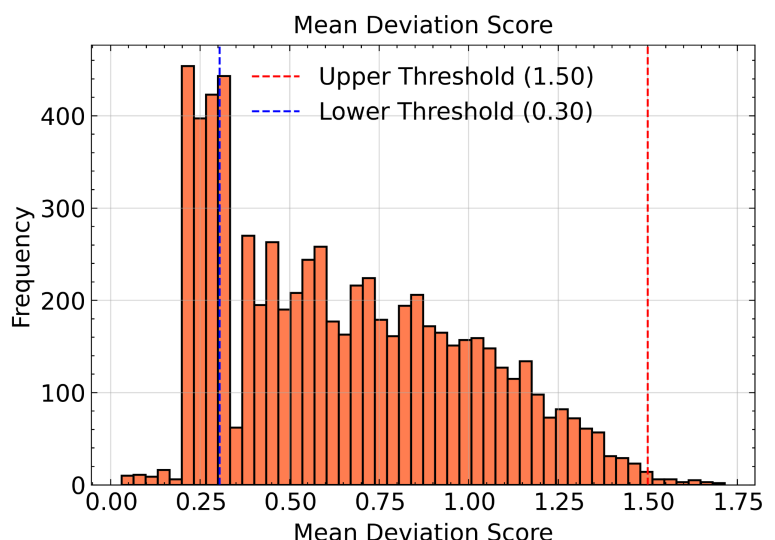


Figure 6: Distribution of Mean Deviation Scores (Iteration 7).

Fig. 5 illustrates the structural topology of the citation network after the hyperbolic embedding and learning process. The visualization reveals a dense core of highly interconnected nodes, representing "normal" academic discourse where citation patterns follow standard community structures driven by homophily. In contrast, peripheral nodes and nodes with irregular connectivity patterns are visually distinct. This spatial organization confirms that the DiskNet model successfully learned the latent geometry of the citation network, clustering "standard" papers while isolating potential anomalies.

Fig.6 presents a histogram of the Mean Deviation Scores calculated for all nodes in Iteration 7. The x-axis represents the deviation score, while the y-axis shows the

node frequency. Vertical dotted lines denote the dynamically calculated thresholds, where the blue line indicates the lower limit and the red line signifies the upper limit. Most nodes fall between the two predefined thresholds and are therefore labeled as exhibiting "normal" citation behavior. Nodes located in the tails of the distribution, specifically those with scores exceeding the upper threshold (to the right of the red line), are flagged as anomalies. This result demonstrates the model's selectivity: rather than flagging large portions of the network as anomalous, it isolates a limited set of nodes that deviate significantly from the learned group dynamics.

To evaluate detection performance, we focused on the model's ability to retrieve the injected synthetic nodes. Since the ground truth is controlled, we assess the detection performance using Recall. A synthetic node correctly assigned an anomaly score above the defined threshold is counted as a True Positive, while a synthetic node that the model failed to detect is counted as a False Negative.

Detection Results	Iteration Number										Avg
	1	2	3	4	5	6	7	8	9	10	
Original Anomalies	879	996	836	1148	1319	850	1043	1020	1073	1036	1020
Original Anomalies (%)	14.07	15.94	13.38	18.37	21.11	13.6	16.69	16.32	17.17	16.58	16.32
Synthetic Anomalies	463	315	466	478	466	351	592	466	465	469	453
Synthetic Anomalies (%)	74.32	50.56	74.8	76.85	74.8	56.34	95.02	74.8	74.64	75.28	72.74

Table 1: Recall-Based Detection Results for Original and Synthetic Anomalies per Iteration

Table 1 details the performance metrics for each independent validation cycle, as defined in Section 3.1.7. Each column corresponds to a distinct synthetic anomaly injection iteration, while the rows report the number and percentage of detected anomalies. Detection performance is measured using Recall, reported under the *Synthetic Anomalies (%)* row. The final column presents the mean Recall across all iterations, serving as a consolidated measure of the model's overall sensitivity.

The results presented in Table 1 indicate relatively consistent Recall values across independent iterations, suggesting that the proposed model exhibits stable detection behavior under varying injection conditions. This consistency implies that the anomaly score, derived from deviations in hyperbolic coordinates, functions as a reliable discriminator, effectively separating injected synthetic anomalies from organically evolving citation patterns. Furthermore, the calculated mean Recall value computed across all iterations serves as an aggregate performance indicator,

supporting the effectiveness of the proposed strategy in retrieving the majority of injected anomalies.

4.2 Conclusions and Future Work

This study sets out to validate the hypothesis that structural anomalies in citation networks are detectable through deviations in latent hyperbolic geometry. As outlined in the initial proposal, we anticipated that synthetic nodes, lacking the natural formation processes of organic research, would fail to embed consistently within the underlying structure of the network. The quantitative results in Subsection 4.1 strongly support this premise.

The model demonstrated consistently high and stable Recall rates across multiple randomized validation iterations. This performance indicates that synthetic nodes consistently exhibit higher radial deviation scores compared to legitimate papers, thereby validating the core assumption that hyperbolic space provides a stronger and more expressive signal for structural irregularities than traditional topological metrics alone.

This project contributes a novel geometry-based approach to identifying citation fraud, which distinguishes itself from standard metadata-driven methods. The results demonstrate that latent hyperbolic embeddings provide a rigorous mathematical framework for anomaly detection. Consequently, this offers a scalable tool for maintaining the integrity of academic databases against structural manipulation.

The primary goal for future research is to transition the framework from controlled synthetic validation to identifying real-world anomalies in operational databases. Further optimization of hyperbolic embedding and dynamic modeling components may significantly enhance computational efficiency when scaling to massive graph structures. In addition, extending the framework to support link prediction could facilitate the discovery of relevant research across disconnected or emerging disciplines. Another promising direction involves using the hyperbolic coordinates to map the early emergence of new scientific trends before they become densely cited. Finally, the framework could be adapted to predict the long-term impact of academic papers based on their radial depth and angular positioning within the latent hyperbolic space.

References

- [1] Eugene, Garfield (1979) "Citation Indexing: Its Theory and Application in Science, Technology, and Humanities." NewYork: Wiley.
- [2] García-Pérez, G., Allard, A., Serrano, M. Á., & Boguñá, M. (2019). Mercator: uncovering faithful hyperbolic embeddings of complex networks. *New Journal of Physics*, 21(12), 123033.
- [3] Li, H., Jiang, H., Ye, D., Wang, Q., Du, L., Zeng, Y., . & Chen, C. (2024). DHGAT: Hyperbolic representation learning on dynamic graphs via attention networks. *Neurocomputing*, 568, 127038.
- [4] Li, R., Wang, H., Piao, J., Liao, Q., & Li, Y. (2024, August). Predicting Long-term Dynamics of Complex Networks via Identifying Skeleton in Hyperbolic Space. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (pp. 1655-1666).
- [5] Zhang, S., Tong, H., Xu, J., & Maciejewski, R. (2019). Graph convolutional networks: A comprehensive review. *Computational Social Networks*, 6(1), 1–23.

Appendix

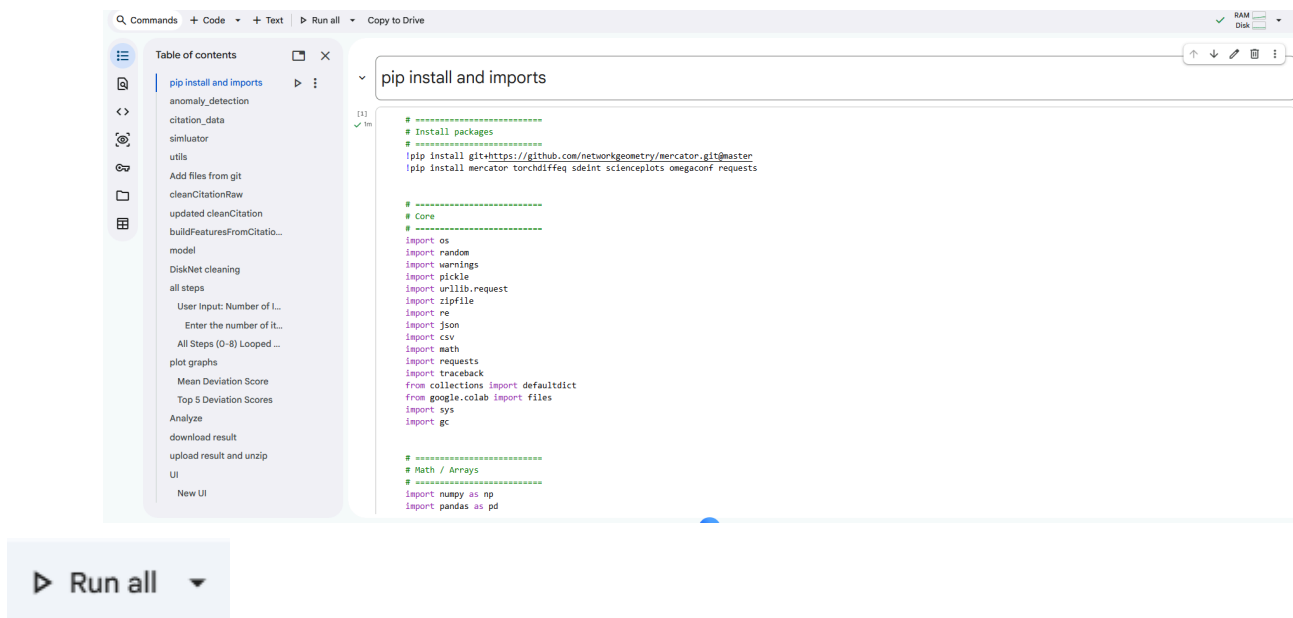
User Guide: DiskNet-Based Anomaly Detection System

Introduction

This document provides a comprehensive user guide for operating the DiskNet-based system designed for anomaly detection in dynamic citation networks. The system is executed via a Google Colab notebook and provides an interactive interface for data upload, processing, analysis, and visualization.

1. Opening and running the System Notebook:

- Open the provided `.ipynb` file using **Google Colab**.
- From the **Runtime** menu, select **"Run All"**.
- Wait until the execution completes.



2. Accessing the User Interface (UI):

Once the execution is complete:

- Scroll to the bottom of the Colab notebook
- Locate the **User Interface (UI)** section.

...

Citation Data Portal

[Go to File Upload](#)

[Check Results](#)

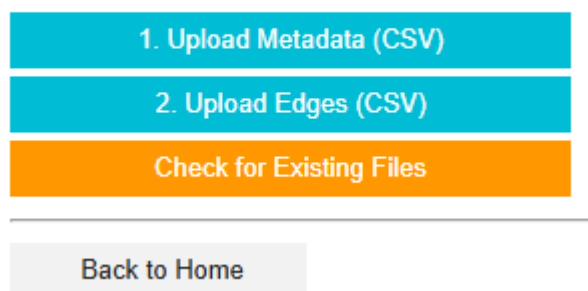
3. Data Upload:

The system requires two input files: a metadata file describing the nodes and a citation edges file describing the relationships between them.

On the main screen, click **"Go to File Upload"** to begin the process.

4. **Uploading Citation Network Data:** Upload the required files according to the specifications:

Step 1: File Upload



1. Upload Metadata (CSV)

2. Upload Edges (CSV)

Check for Existing Files

Back to Home

4.1 Metadata File Requirements:

Upload a metadata file containing the following mandatory columns:

- **id**: Unique identifier for the paper.
- **year**: Publication year.
- **concepts**: Research concepts/tags.
- **primary_topic**: The main topic of the paper.

Important constraints:

- The publication years must be **continuous, with** no gaps.
- The dataset must span **exactly 21 consecutive years**.

Step 1: File Upload

1. Upload Metadata (CSV)

2. Upload Edges (CSV)

Check for Existing Files

[Back to Home](#)

Please select your meta file...

No file chosen

Step 1: File Upload

1. Upload Metadata (CSV)

2. Upload Edges (CSV)

Check for Existing Files

[Back to Home](#)

Please select your meta file...

No file chosen

Upload widget is only available

Saving papers_metadata.csv to papers_metadata.csv

✔ Metadata Validated and Saved: Metadata Validated

4.2 Citation Edges File Requirements:

Upload an edges file describing citation relationships with the following columns:

- **cites**: The ID of the citing paper.
- **cited**: The ID of the paper being cited.

Step 1: File Upload

1. Upload Metadata (CSV)

2. Upload Edges (CSV)

Check for Existing Files

[Back to Home](#)

Please select your edges file...

 No file chosen

Step 1: File Upload

1. Upload Metadata (CSV)

2. Upload Edges (CSV)

Check for Existing Files

5. Data Cleaning and Processing:

After uploading the files, the system automatically navigates to the data cleaning stage.

Step 2: Data Cleaning

Clean & Process Data

Back

Click the **"Clean & Process Data"** button to initiate preprocessing.

Step 2: Data Cleaning

Clean & Process Data

Back

⌚ Starting full data cleaning process...
Removed self-citations: 224 rows dropped
After dropping duplicate edges: 24814 rows
Unique IDs appearing in edges: 6503
Removed isolated articles: 713 rows dropped
Filtered edges to valid meta IDs: 0 rows dropped
Corrected citation direction based on year: 22809 edges swapped
Final edges count: 24763
Saved cleaned edges to: /content/data/citation/clean_edges.csv
Saved cleaned meta to: /content/data/citation/clean_meta.csv
Saved /content/data/citation/id2idx_clean.json (N=6503)
✅ Data cleaning complete!

6. Model Execution:

- The system proceeds automatically to the model execution phase.
- Processing may take several minutes, depending on the dataset size
- Upon completion, the system redirects to the analysis screen.

Step 3: Ready for Analysis

[Back](#)[Start Analysis](#)

Step 4: Pipeline Execution

Running 1 iterations...

▶ [Run 1/1] Initializing...

```
=====
Starting Run 1/1
=====
Data directory for this run: /content/data/citation_run_1
Log directory for this run: logs/citation_run_1/l3_h3/CitationNetwork/DiskNet

=====
STEP 0: Synthetic Citation Network Generator (Iteration-specific)
=====
Loading data files...
Loaded clean_meta.csv: 6503 rows
Loaded clean_edges.csv: 24763 rows
Loaded topics and concepts: 7 topics

Applying pre-cleaning logic to original graph data...
Initial clean graph: 6503 nodes, 24763 edges
After removing isolates: 6503 nodes
After removing self-loops: 24763 edges
After keeping LWCC: 6250 nodes, 24584 edges
Pre-cleaning complete. Final original graph size: 6250 nodes, 24584 edges

Calculating network parameters...

Creating synthetic nodes...
Created 623 synthetic nodes
```

7. Analysis and Visualization:

On the analysis screen, you can review the processed results and visualizations:

Step 5: Analysis & Results

[Analyze Summary](#)[Generate Plots](#)[Back to Home](#)

7.1 Results Analysis:

Review anomaly scores and evaluation metrics produced by the model.

Step 5: Analysis & Results

Analyze SummaryGenerate Plots

Run	Total Nodes	Anomalies	Synth Recall %	Original Anom %	
0	1	6872	1341	74.317817	14.050248

Back to Home

7.2 Visualization:

Select your preferred graph visualization from the menu to explore the detected anomalies.

Step 5: Analysis & Results

Analyze SummaryGenerate Plots

✓ All plots regenerated with full labels and thresholds.

Select Plot Folder:

Back to Home

Analyze SummaryGenerate Plots

Select Plot Folder:

Showing all plots from: top_5_deviation_scores

File: run_1_deviation_scores.png

Run 1 - Top 5 Extreme Nodes Deviation Timeline

Deviation Score

Test Window Index

Node W2127774996
Node W2138644576
Node W2096525273
Node W2142831770
Node W2159675211

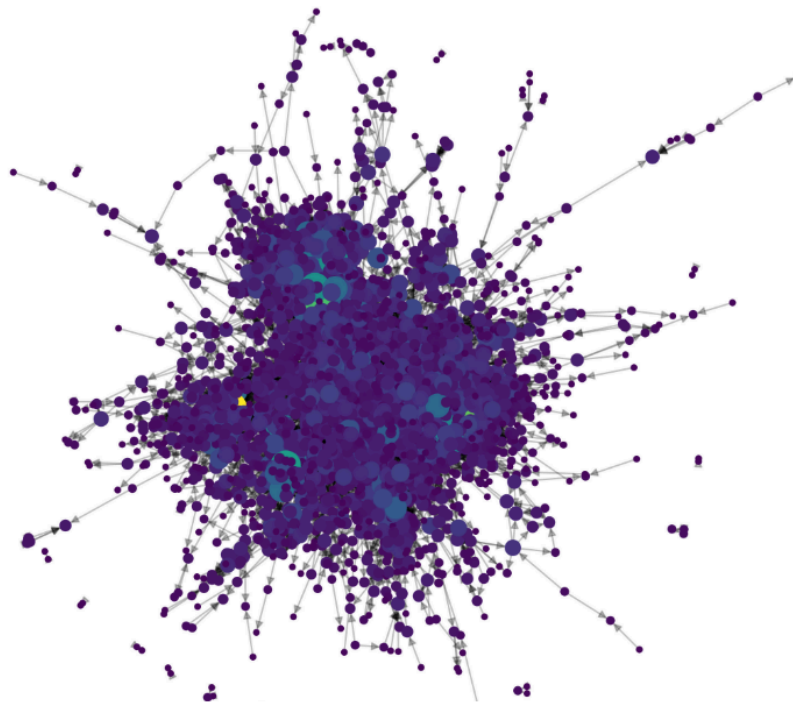
Test Window Index	Node W2127774996	Node W2138644576	Node W2096525273	Node W2142831770	Node W2159675211
0.0	2.4	1.7	2.4	1.7	1.7
1.0	1.4	2.5	1.8	2.4	1.8
2.0	1.5	1.4	1.7	1.2	1.5
3.0	1.6	1.5	1.5	2.0	1.4
4.0	1.9	1.7	1.3	1.3	2.1

Back to Home

Showing all plots from: **graph**



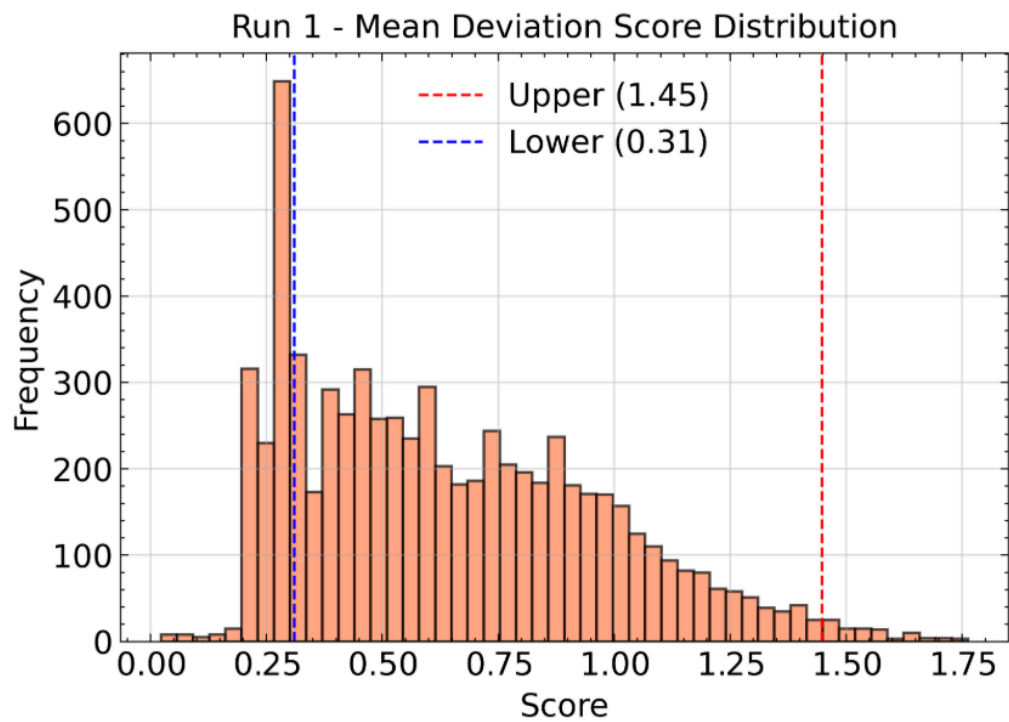
File: run_1_graph.png



Showing all plots from: **Mean Deviation Score**



File: run_1_mean_scores_hist.png



Maintenance Guide: DiskNet for Citation Networks

1. System Environment

This section describes the necessary software infrastructure to run and maintain the project.

Software Requirements

- **Language:** Python 3.8 or higher

Core Libraries:

- **torch** (PyTorch) - For the DiskNet model and neural network operations.
- **networkx** - For graph manipulation and network analysis.
- **pandas & numpy** - For data processing and numerical calculations.
- **scikit-learn** - Specifically for the **IsolationForest** anomaly detection.
- **mercator** - External tool for hyperbolic embedding (installed via GitHub).

Installation Instructions

To set up the environment, run the following commands in your Python environment:

```
# Install core dependencies
pip install torch networkx pandas numpy scikit-learn matplotlib

# Install the Mercator library for hyperbolic mapping
pip install git+https://github.com/networkgeometry/mercator.git@master
```

2. Key Maintenance & Customization Points

For future research or adaptation to different datasets, the following modules and parameters are the most critical to modify:

A. Feature Engineering (**build_features_from_citation_csv**)

- The system's performance heavily relies on how node dynamics are represented.
- **Modifying Dynamics:** Currently, it extracts citation-based counts over time. It can be adapted to any time-series data associated with nodes and change the type of dynamics analyzed by modifying the logic within this function.
- **Quantity of Dynamics:** You can increase or decrease the number of features per node. If you change the input vector size, ensure the **input_dim** in the model configuration is updated accordingly to match the new feature set.

✓ buildFeaturesFromCitationCsv

```
[9] def build_dynamics_from_graph(G, meta_csv, out_dir):  
    os.makedirs(out_dir, exist_ok=True)  
  
    # 1. LOAD METADATA  
    meta_df = pd.read_csv(meta_csv, dtype=str)  
    meta_df.columns = [c.strip() for c in meta_df.columns]  
    meta_df["id"] = meta_df["id"].astype(str)  
  
    # 2. LOAD GRAPH ID MAPPINGS  
    graph_map_path = os.path.join(out_dir, 'graph_node_map_int_to_str.json')  
    with open(graph_map_path, "r") as f:  
        int_to_orig_id_map = {int(k): v for k, v in json.load(f).items()}  
        orig_id_to_int_map = {v: k for k, v in int_to_orig_id_map.items()}  
  
    # 3. PREPARE METADATA ARRAYS  
    N = G.number_of_nodes()  
    id2idx = orig_id_to_int_map  
    years = np.full(N, -1, dtype=int)  
    primary_topics = [None] * N  
  
    for _, row in meta_df.iterrows():  
        if row["id"] in id2idx:  
            idx = id2idx[row["id"]]  
            try:  
                years[idx] = int(row["year"])  
            except:  
                years[idx] = -1  
            primary_topics[idx] = row.get("primary_topic", None)  
  
    # 4. PRE-CALCULATE GRAPH DISTANCES (for f4)  
    # We use an undirected version of the graph to find 'community distance'  
    G_undirected = G.to_undirected()
```

B. Global Configuration (**Config** class)

The **Config** class centralizes all hyperparameters for the time-series forecasting. Adjust these to tune the model's "vision":

- **Lookback:** Determines how many past time steps the model considers for prediction.

- **Horizon:** Determines how many steps into the future the model attempts to predict.
- **Data Ratios:** The `train_ratio`, `val_ratio`, and `test_ratio` control the data split. Adjust these if the dataset size changes significantly to ensure the model has enough samples to learn effectively.

```
# Fixed Parameters
L_FIXED = 3
H_FIXED = 3
K_FIXED = 1
TRAIN_RATIO_FIXED = 0.6
VAL_RATIO_FIXED = 0.15
```

C. Synthetic Anomaly Generation

The system evaluates its detection capabilities by injecting a synthetic network into the original data.

- **Network Size:** Controlled by the ratio of synthetic nodes (default: 10% of the total network). Increasing this will make the anomaly detection task more prominent but potentially less realistic.
- **Connectivity:** Density relative to the original graph (default: 30% of the original network's density). This ratio dictates how "hidden" or "obvious" the synthetic cluster is within the global graph structure. Adjusting this parameter allows you to test the model's sensitivity to different structural anomalies.

```
def calculate_network_parameters(clean_meta_df: pd.DataFrame, clean_edges_df: pd.DataFrame):
    print("\nCalculating network parameters...")
    original_nodes = len(clean_meta_df)
    new_nodes_total = math.floor(original_nodes * 0.1)
    original_edges = len(clean_edges_df)
    new_edges_total = math.floor(original_edges * 0.3)
    return new_nodes_total, new_edges_total
```

3. Maintenance Procedures

1. **Retraining:** Full retraining of the DiskNet model is required after any significant change to `Lookback`, feature definitions, or `input_dim` (input dimensions).
2. **Threshold Tuning:** The `contamination` parameter in the `CitationAnomalyDetector` should be adjusted based on the expected anomaly ratio in the specific real-world use case.